higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1: INTRODUCTION TO FORMWORK AND FALSEWORK

1.1 1.1.1 Formwork is a temporary structure✓ or mould into which new concrete is placed✓ and compacted to harden.✓

1.1.2 Falsework is any temporary structure✓ used to support a permanent structure✓ (under construction) while it is not self-supporting until it is stable enough to support itself.✓

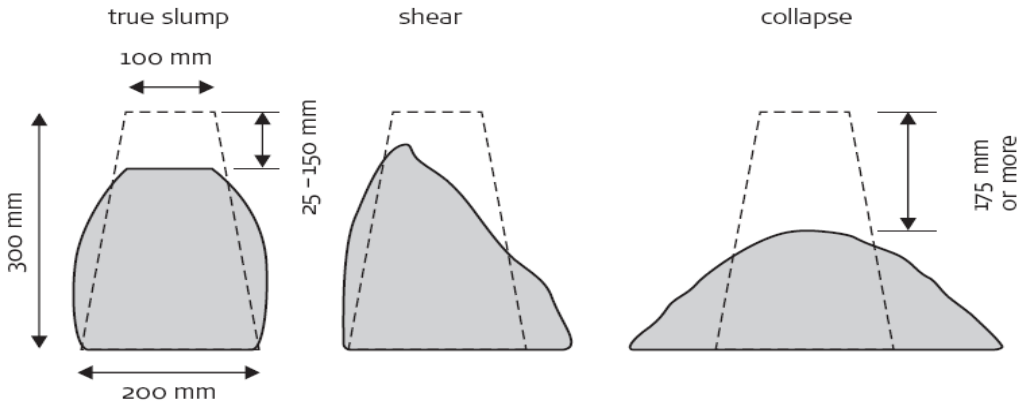
(2 × 3) (6)

1.2
- Strength
- Stiffness
- Finish and texture
- Contamination
- Stripping and striking of formwork
- Reusability

(Any 4 × 1) (4)

[10]

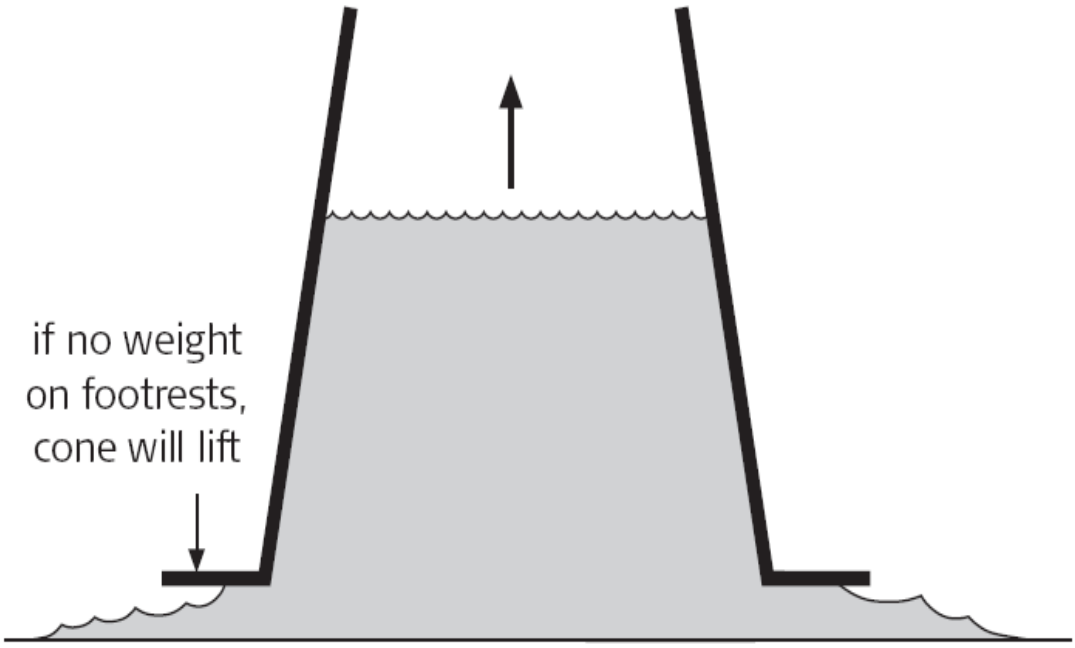
QUESTION 2: FORM FACE MATERIALS AND SPECIAL TYPES OF FORMS



Labels: 5
Correctness: 5 **[10]**

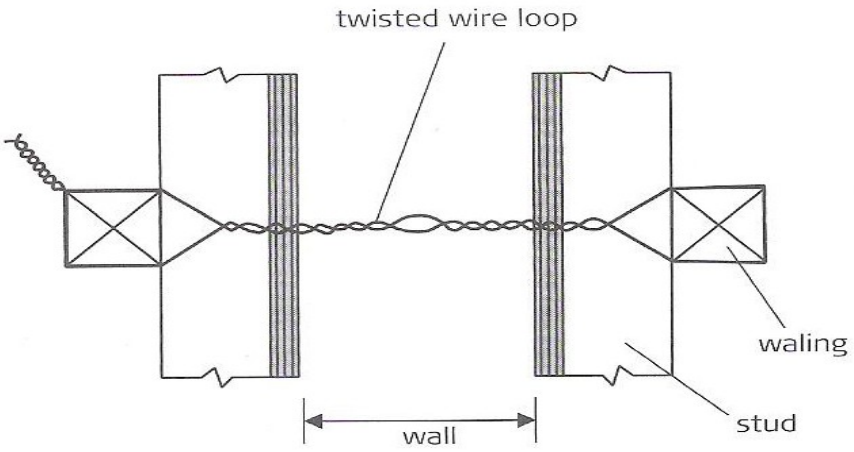
QUESTION 3: PRESSURE RESISTED BY FORMWORK AND FALSEWORK

3.1



Labels: 3
Correctness: 4 (7)

3.2



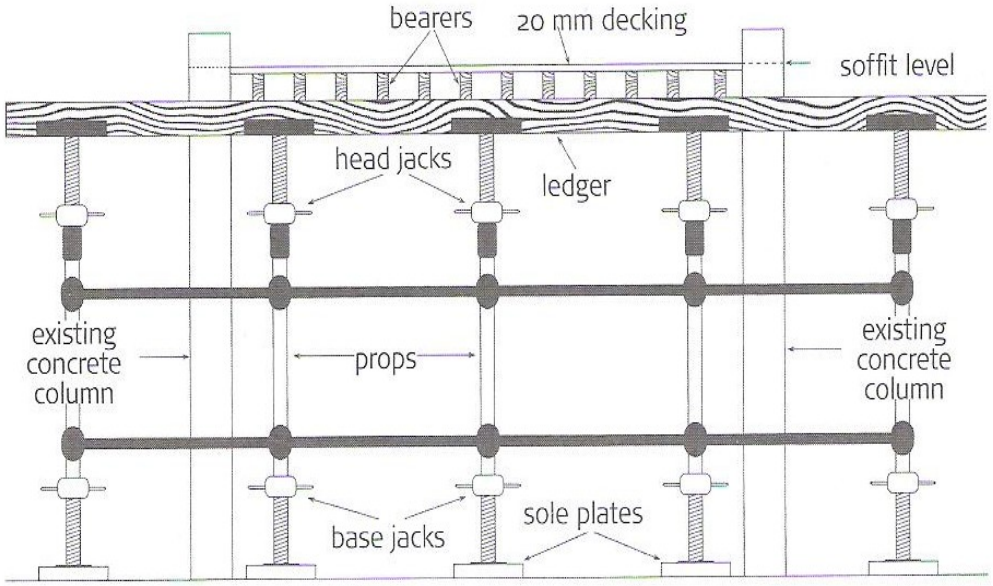
Labels: 3
Correctness: 1 (4)

3.3
- Density of the concrete
- Workability of the concrete
- Rate of placing
- Method of discharge
- Concrete temperature (Any 4 × 1) (4)

[15]

QUESTION 4: ERECTING FORMWORK AND FALSEWORK

4.1



Labels: 8
Correctness: 4 (12)

4.2 4.2.1 The long horizontal lengths of timber or metal used at the top of falsework for formwork

4.2.2 The shorter lengths of timber or metal placed across the ledgers used for falsework

4.2.3 The shorter lengths of timber or metal placed across the ledgers used for scaffolding

4.2.4 Stronger beams usually made of wood that companies manufacture especially for formwork

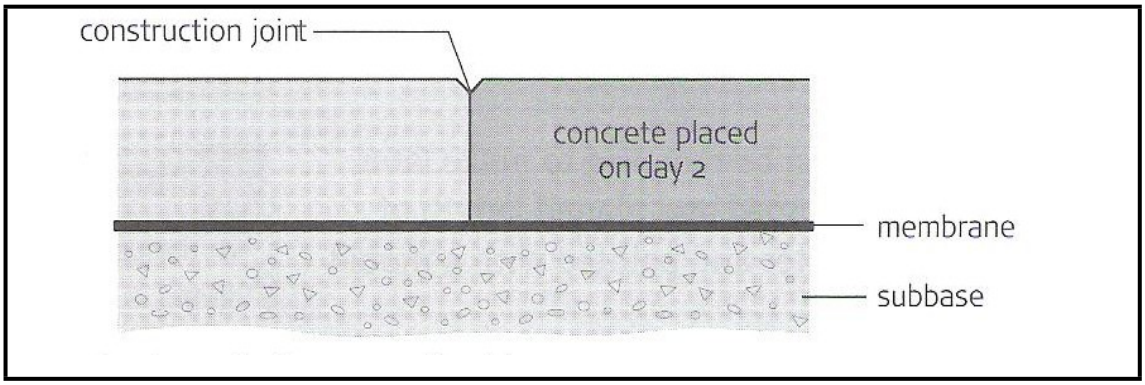
(4 × 2) (8)

4.3
- Join panels together with other panels
- Put together column boxes
- Join wallings to panels
- Join column boxes to panels
- Join soldiers to panels and wallings (5)

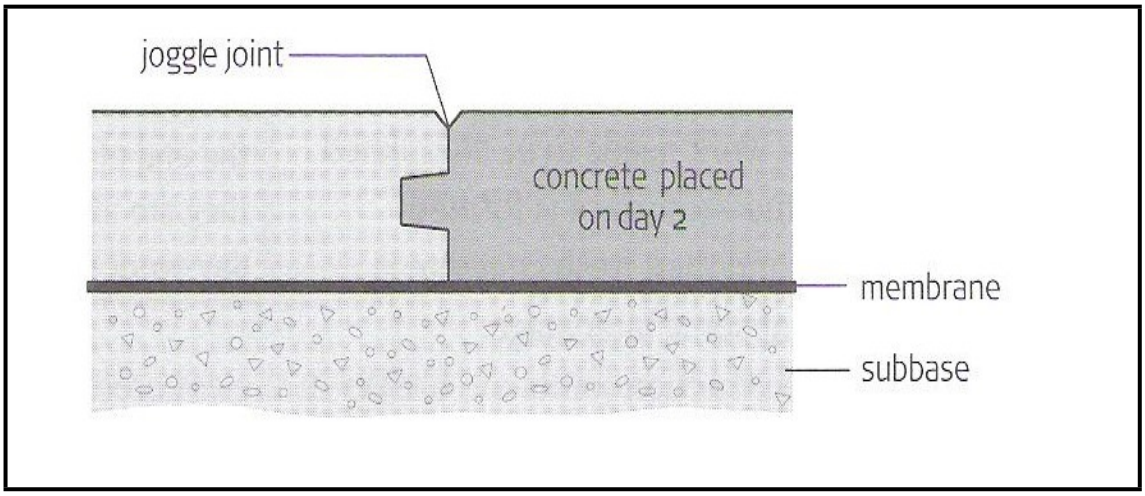
[25]

QUESTION 5: VERTICAL STOP ENDS, WATER AND JOINTS

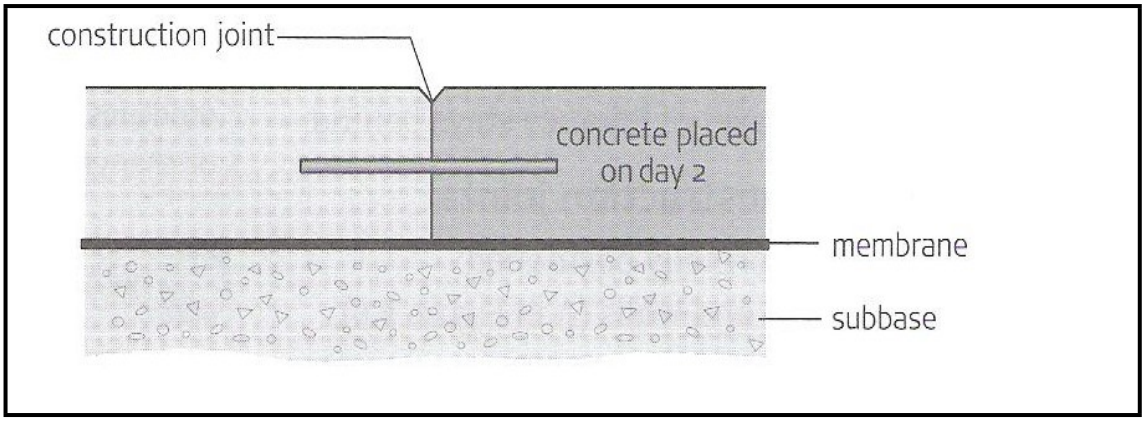
5.1



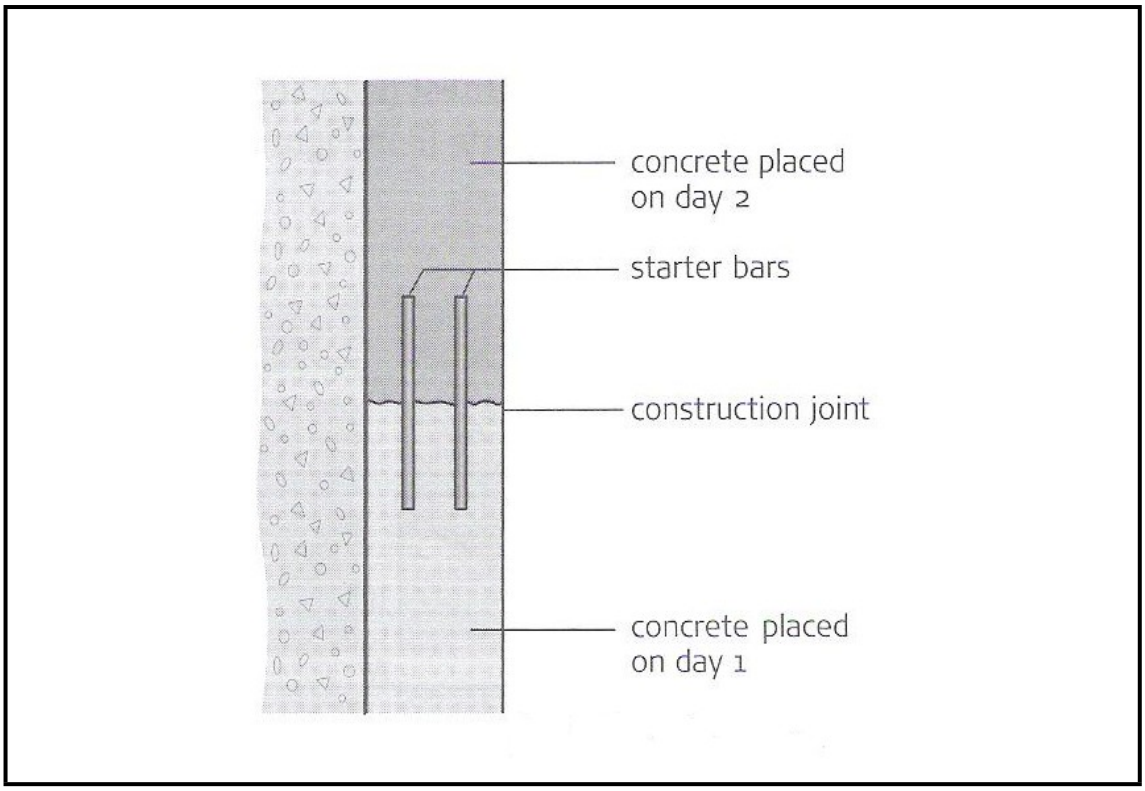
5.2



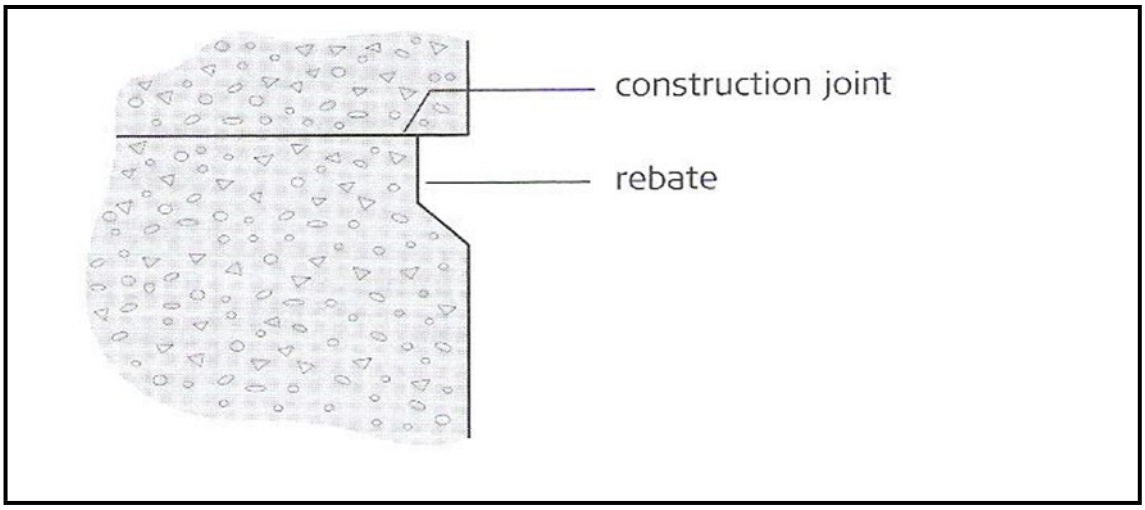
5.3



5.4


concrete placed on day 2
starter bars
construction joint
concrete placed on day 1

5.5


construction joint
rebate

(5 × 2) **[10]**

QUESTION 6: USING RELEASING AGENTS

6.1
- Always follow the manufacturer's instructions.
- Never water down or mix releasing agents with other materials.
- Never store releasing agents in extreme temperatures.
- Avoid contact with skin, eyes or clothing. (4)

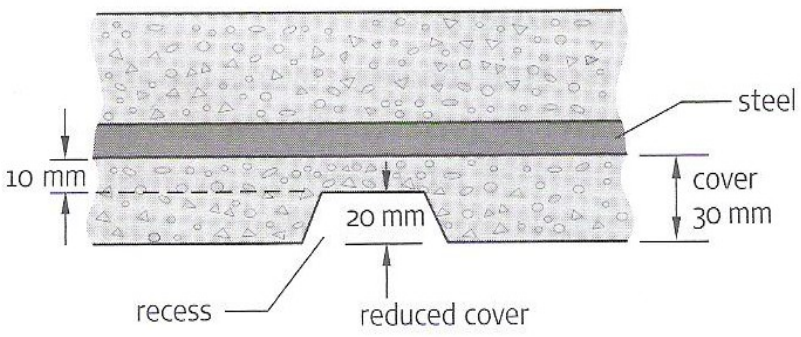
6.2
- Each time the formwork is to be used
- Before fixing reinforcement
- By the method and at the coverage recommended by the manufacturer (3)

6.3 Chemical releasing agents are made from a small amount of chemicals suspended in petroleum distillate.✓ When applied to a form face, the petroleum distillate evaporates,✓ leaving a film on the surface.✓ (3)

[10]

QUESTION 7: WORKMANSHIP AND DETAILING IN CONCRETE STRUCTURES

7.1



Labels: 6
Correctness: 2 (8)

7.2
- The geometric dimensions of the formwork should be given.
- The accuracy of horizontal surfaces should be controlled by levels and the vertical surfaces by plumb lines.
- All forms must be perfectly clean.
- All joints must be flush and tight, and sealed as required.
- There should be no nails that will stick out into the concrete.
- Water stops must be fixed and positioned correctly.
- The correct number of ties should be used.
- Ties must be of the correct length and must be fitted with proper washers.
- Ties must be tight enough, but not too tight.
- All releasing agents must be applied correctly. (10)

7.3
- Normal deterioration of concrete
- Using incorrect tolerances (2)

[20]

TOTAL: 100

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